

# Multi-Agent LLM Framework for 5G Network Slice Resource Management

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Mid-Term Review

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# Outline

Background & Motivation

Methodology

Experimental Results

Problems, Solutions & Future Work

# Research Background · Existing Challenges · Motivation

**5G Network Slicing** creates multiple logical networks over shared infrastructure. Three slice types share limited spectrum with **competing** QoS (Quality of Service) demands:

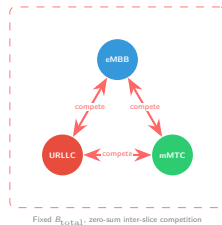
**eMBB** (high throughput) / **URLLC** (ultra-low latency) / **mMTC** (massive access). Traditional rule-based and DRL (Deep Reinforcement Learning) methods struggle to unify 3GPP SLA (Service Level Agreement) constraints, operator policies, and dynamic user behaviour into an adaptive framework.

**eMBB** (Enhanced Mobile Broadband): high-throughput services, e.g. video streaming.

**URLLC** (Ultra-Reliable Low-Latency Comm.): mission-critical apps, sub-ms latency.

**mMTC** (Massive Machine-Type Comm.): massive IoT device connectivity.

| Framework                 | Architecture            | Multi-Agent | Negotiation          | E2E Slicing |
|---------------------------|-------------------------|-------------|----------------------|-------------|
| WirelessAgent [1]         | Percept-Memory-Plan-Act | ✗           | ✗                    | ✗           |
| CommLLM [2]               | Three-component MAS     | ✓           | ✗                    | ✗ Semantic  |
| Qu et al. [3]             | Dual-loop edge-terminal | ✓           | Centralised          | Partial     |
| MetaGPT [4] / ChatDev [5] | Role-based cooperative  | ✓           | ✓ Cooperative        | ✗ General   |
| <b>This Work</b>          | <b>Competitive MAS</b>  | ✓           | ✓ <b>Competitive</b> | ✓           |



## Description

**Gap 1** End-to-end multi-agent LLM (Large Language Model) framework for slicing remains underexplored

**Gap 2** Competitive negotiation for constrained resource allocation is unaddressed by cooperative MAS (Multi-Agent System)

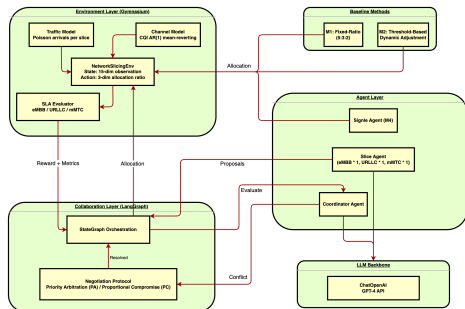
**RQ1** Can multi-agent LLM outperform rule-based and single-agent baselines?

**RQ2** Does negotiation provide structural advantages over direct allocation?

# Methodology: Architecture · Two-Tier Scheduling · Negotiation

**System Architecture** — Agent Layer / Collaboration Layer (LangGraph) / Environment Layer

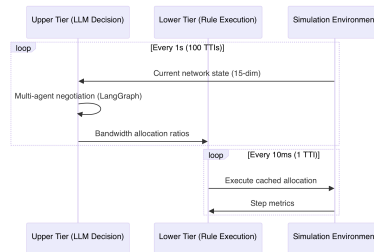
(Gymnasium):



**Negotiation Flow (LangGraph StateGraph, max 3 rounds):**



**Two-Tier Scheduling** — Bridging LLM inference latency (seconds) vs network real-time requirements (milliseconds):



| ID    | Method           | Category   | Description                         |
|-------|------------------|------------|-------------------------------------|
| M1    | Fixed-Ratio      | Rule-based | Fixed 5:3:2 allocation              |
| M2    | Threshold        | Rule-based | SLA-triggered dynamic adjustment    |
| M4    | Single-Agent LLM | Single LLM | GPT-4 + CoT (Chain-of-Thought)      |
| M5-PA | Multi-Agent PA   | Multi LLM  | Priority arbitration negotiation    |
| M5-PC | Multi-Agent PC   | Multi LLM  | Proportional compromise negotiation |

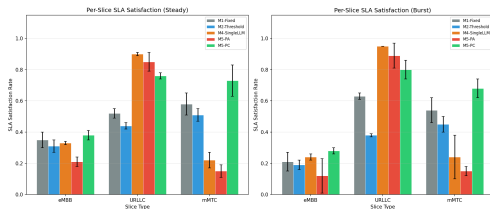
# Experiment 1: Core Performance Comparison (RQ1)

**Steady-State Scenario** (100 users, 100 decision cycles  $\times$  2 seeds):

| Method       | $SLA_{eMBB}$    | $SLA_{URLLC}$                     | $SLA_{mMTC}$                      | $\bar{S}$   | Fairness     | Lat. (ms) | Tokens |
|--------------|-----------------|-----------------------------------|-----------------------------------|-------------|--------------|-----------|--------|
| M1-Fixed     | $0.35 \pm 0.05$ | $0.52 \pm 0.03$                   | $0.58 \pm 0.07$                   | 0.48        | 0.893        | 0         | 0      |
| M2-Threshold | $0.31 \pm 0.04$ | $0.44 \pm 0.02$                   | $0.51 \pm 0.04$                   | 0.42        | 0.912        | 0         | 0      |
| M4-SingleLLM | $0.33 \pm 0.01$ | <b><math>0.90 \pm 0.01</math></b> | $0.22 \pm 0.05$                   | 0.48        | 0.720        | 6326      | 26k    |
| M5-PA        | $0.21 \pm 0.03$ | $0.85 \pm 0.06$                   | $0.15 \pm 0.04$                   | 0.40        | 0.616        | 8065      | 125k   |
| M5-PC        | $0.38 \pm 0.03$ | $0.76 \pm 0.02$                   | <b><math>0.73 \pm 0.10</math></b> | <b>0.62</b> | <b>0.952</b> | 8033      | 125k   |

**Burst Scenario** (eMBB users doubled at cycle 50):

| Method       | $SLA_{eMBB}$    | $SLA_{URLLC}$                     | $SLA_{mMTC}$                      | $\bar{S}$   | Fairness     | Lat. (ms) | Tokens |
|--------------|-----------------|-----------------------------------|-----------------------------------|-------------|--------------|-----------|--------|
| M1-Fixed     | $0.21 \pm 0.06$ | $0.63 \pm 0.02$                   | $0.54 \pm 0.08$                   | 0.46        | 0.831        | 0         | 0      |
| M2-Threshold | $0.19 \pm 0.03$ | $0.38 \pm 0.01$                   | $0.45 \pm 0.05$                   | 0.34        | 0.876        | 0         | 0      |
| M4-SingleLLM | $0.24 \pm 0.02$ | <b><math>0.95 \pm 0.00</math></b> | $0.24 \pm 0.14$                   | 0.48        | 0.660        | 6063      | 26k    |
| M5-PA        | $0.12 \pm 0.11$ | $0.89 \pm 0.08$                   | $0.15 \pm 0.03$                   | 0.38        | 0.541        | 7957      | 125k   |
| M5-PC        | $0.28 \pm 0.02$ | $0.80 \pm 0.06$                   | <b><math>0.68 \pm 0.06</math></b> | <b>0.59</b> | <b>0.924</b> | 7887      | 125k   |

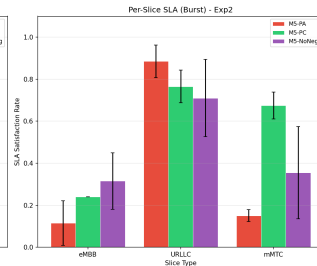
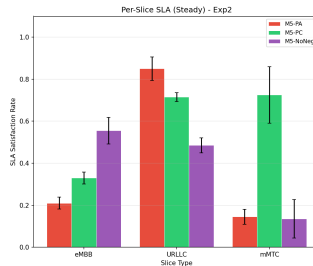
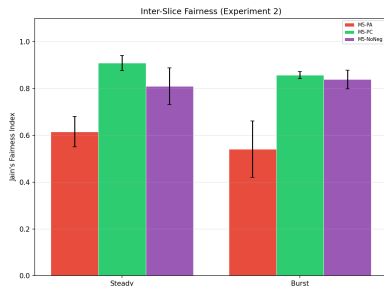


## Key Findings (RQ1):

- M5-PC outperforms all baselines in overall SLA ( $\bar{S}=0.62$ ) with Fairness=0.952, far above M4 (0.720) and M5-PA (0.616)
- M5-PA over-prioritises URLLC at the expense of eMBB/mMTC
- M4 tends toward extreme allocation: very high URLLC (0.90) but very low mMTC (0.22)
- Rule methods have zero latency but cannot reason about SLA semantics
- **Proportional compromise is the best strategy for balancing SLA and fairness**

## Experiment 2: Negotiation Mechanism Validation (RQ2)

| Method   | Scenario | SLA <sub>eMBB</sub> | SLA <sub>URLLC</sub> | SLA <sub>mMTC</sub> | $\bar{S}$   | Fairness     |
|----------|----------|---------------------|----------------------|---------------------|-------------|--------------|
| M5-PA    | steady   | 0.21                | 0.85                 | 0.15                | 0.40        | 0.616        |
| M5-PA    | burst    | 0.12                | 0.89                 | 0.15                | 0.38        | 0.541        |
| M5-PC    | steady   | 0.38                | 0.76                 | 0.73                | <b>0.62</b> | <b>0.952</b> |
| M5-PC    | burst    | 0.28                | 0.80                 | 0.68                | <b>0.59</b> | <b>0.924</b> |
| M5-NoNeg | steady   | 0.56                | 0.49                 | 0.14                | 0.39        | 0.810        |
| M5-NoNeg | burst    | 0.32                | 0.71                 | 0.36                | 0.46        | 0.839        |



### Key Findings (RQ2) — Structural Advantage of Negotiation:

- **M5-PC vs M5-NoNeg (steady):**  $\bar{S} + 59\%$  ( $0.39 \rightarrow 0.62$ ), Fairness  $+18\%$  ( $0.81 \rightarrow 0.95$ )
- **M5-PC vs M5-PA:** Proportional compromise outperforms priority arbitration across all metrics, especially mMTC SLA ( $0.73$  vs  $0.15$ )
- **M5-NoNeg** uses LLM but lacks iterative negotiation, resulting in imbalanced allocation (eMBB=0.56, mMTC=0.14)
- **M5-PA** hard priority causes URLLC to monopolise resources ( $0.85+$ ); low-priority slices are systematically under-served
- **Conclusion:** The negotiation protocol itself (not merely more LLM calls) drives significant SLA and fairness improvements

# Problems & Solutions · Future Work

## Identified Problems

- **MAS latency incompatible with real-time control.** Multi-agent negotiation takes  $\sim 8$  s per step, far exceeding the millisecond-level response time required by network control loops.
- **Statistical significance.** Current experiments use only 2 random seeds with 100 decision cycles per episode, which limits the robustness of conclusions.

## Proposed Solutions

- **Strategic MAS + tactical rule executor** (*in progress*): Reposition MAS as a **strategic layer** that predicts traffic trends and generates allocation *policies*, rather than making per-TTI decisions. The lower-tier rule executor operates at 10 ms granularity guided by MAS-provided policies, decoupling LLM inference from the real-time control loop.
- **Expanded experimental scale:** Increase to 5 random seeds  $\times$  500 decision cycles per episode to strengthen statistical validity.

## Future Work

- **Strategic MAS + tactical rule executor:** Validate the two-tier paradigm where MAS provides medium-term allocation policies and the rule layer handles real-time execution
- **DRL baseline (M3-PPO):** Train PPO (Proximal Policy Optimisation) with Stable-Baselines3 as an additional baseline
- **RAG (Retrieval-Augmented Generation) integration:** Inject 3GPP specifications and telecom domain knowledge into agent prompts
- **Agent memory:** Leverage historical allocation patterns to improve decision quality over time

**Setup:** 100 MHz (NR n78), 100 users, GPT-4o-mini (temp=0), negotiation  $\leq 3$  rounds, Gymnasium + LangGraph.

## References

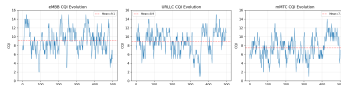
- [1] Jingwen Tong et al. “WirelessAgent: Large Language Model Agents for Intelligent Wireless Networks”. In: *arXiv preprint arXiv:2505.01074* (2025).
- [2] Feibo Jiang et al. “Large Language Model Enhanced Multi-Agent Systems for 6G Communications”. In: *IEEE Wireless Communications* 31.6 (2024), pp. 48–55.
- [3] Zhiyuan Qu et al. “LLM Enabled Multi-Agent System for 6G Networks: Framework and Method of Dual-Loop Edge-Terminal Collaboration”. In: *arXiv preprint arXiv:2509.04993* (2025).
- [4] Sirui Hong et al. “MetaGPT: Meta Programming for Multi-Agent Collaborative Framework”. In: *arXiv preprint arXiv:2308.00352* (2023).
- [5] Chen Qian et al. “ChatDev: Communicative Agents for Software Development”. In: *arXiv preprint arXiv:2307.07924* (2023).

**Thank You!**

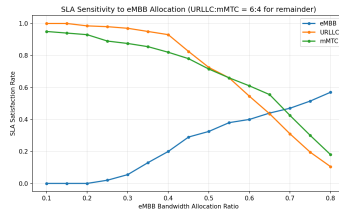
# Appendix A: Simulation Environment Validation



Poisson arrival validation ( $\chi^2$  test)



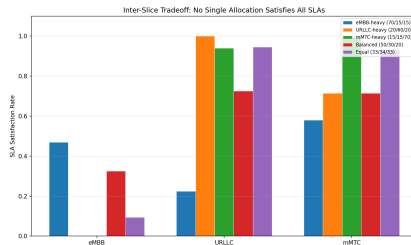
CQI AR(1) mean-reverting evolution



SLA sensitivity to allocation



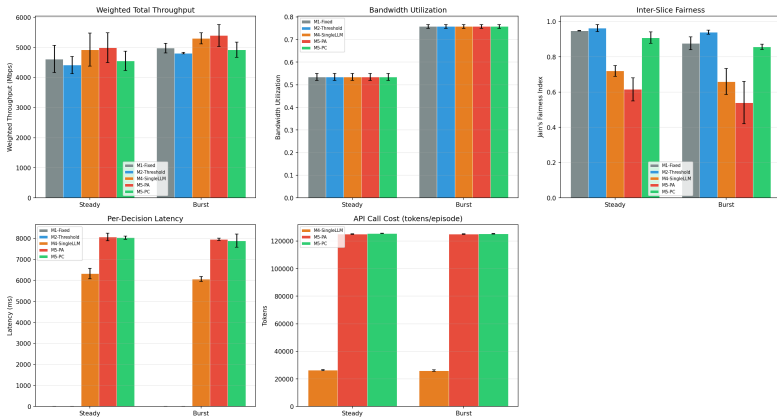
Burst event impact validation



Inter-slice resource competition tradeoff

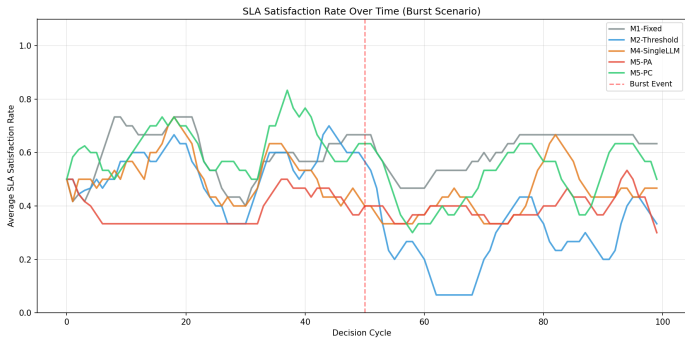
## Appendix B: Experiment 1 — Comprehensive Performance Overview

Experiment 1: Comprehensive Performance Comparison



Throughput, bandwidth utilisation, fairness, decision latency, and API cost across all five methods

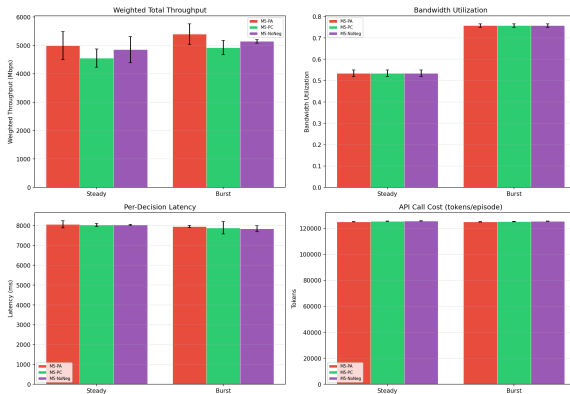
## Appendix C: Experiment 1 — Burst Scenario SLA Trend



SLA satisfaction rate over decision cycles under burst scenario (eMBB users doubled at cycle 50)

## Appendix D: Experiment 2 — Comprehensive Performance Overview

Experiment 2: Negotiation Mechanism Performance



M5-PA / M5-PC / M5-NoNeg throughput, utilisation, latency, and API cost comparison